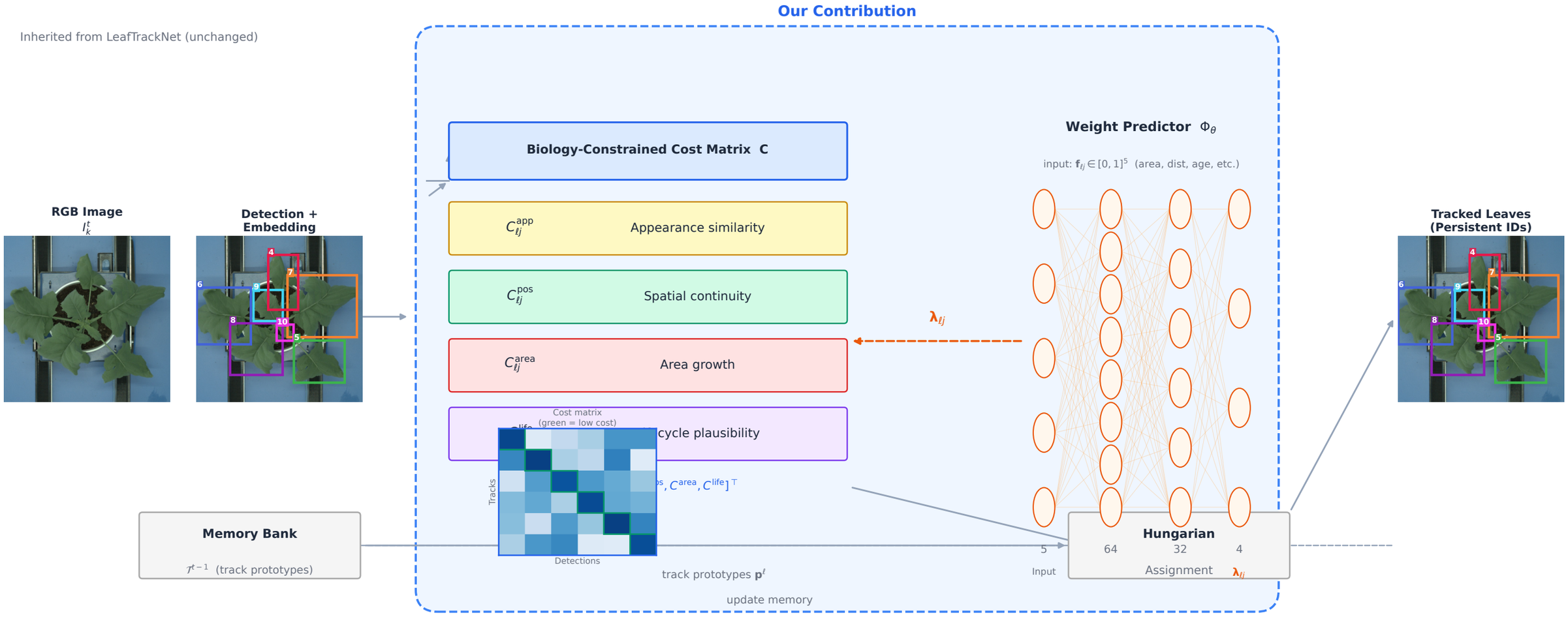
Our Contribution
Inherited from LeafTrackNet (unchanged)
RGB Image
I_k^t
Detection + Embedding
Biology-Constrained Cost Matrix C
$C_{\ell j}^{\text{app}}$ Appearance similarity
$C_{\ell j}^{\text{pos}}$ Spatial continuity
$C_{\ell j}^{\text{area}}$ Area growth
cycle plausibility
Cost matrix (green = low cost)
Tracks
Detections
Memory Bank
$\mathcal{T}^{t-1}$ (track prototypes)
track prototypes $\mathbf{p}^\ell$
update memory
Weight Predictor Φ_θ
input: $\mathbf{f}_{\ell j} \in [0,1]^5$ (area, dist, age, etc.)
$\lambda_{\ell j}$
5
64
32
4
Input
Hungarian
Assignment
$\lambda_{\ell j}$
Tracked Leaves (Persistent IDs)